

JAYANTH DASAMANTHARAO

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PROFESSIONAL EXPERIENCE

Applied AI/ML Engineer-Fedway Associates, EliteUS Software Solutions Aug 2024 - Present | Edison, NJ

- Translated business and product needs into production ML systems, designing optimization-driven models over 5M+ interaction records to improve relevance, efficiency, and user engagement by 30%.
- Designed and built production ML pipelines for targeting, relevance, and allocation optimization, leveraging embeddings and retrieval-based features over large-scale datasets.
- Designed and executed rigorous experimentation frameworks, including A/B and lift tests to measure incrementality and model impact on key business and marketplace metrics, iterating based on telemetry.
- Owned production reliability for ML services, building CI/CD pipelines, automated testing, monitoring, and alerting for real-time ML APIs at scale.

Data Science Intern, Harvest Software Solutions June 2023 - Sept 2023 | Jacksonville, Florida

- Engineered scalable ML pipelines and monitoring systems for large advertising datasets, increasing campaign visibility from 15% to 45% while ensuring reliability and performance.
- Built ML services and NLP APIs using XGBoost, H2O AutoML, and FastAPI, improving sentiment classification accuracy by 65% and campaign spend efficiency by 50%.
- Designed SQL/Python pipelines for feature engineering, data modeling, and model training, supporting experimentation and performance optimization for large-scale datasets.

Application Development Associate, Accenture June 2021 - Aug 2022 | Hyderabad, India

- Developed and optimized Python-based ETL and data processing pipelines using hash maps, sets, and indexed lookups to efficiently clean, deduplicate, transform large datasets, reducing 50% production issues.
- Implemented SQL workflows and joins with partitioning, indexing, and window functions, enabling efficient aggregation and analysis over millions of records in AWS Athena and Redshift.
- Applied batching, caching, and algorithmic optimizations for data validation and pipeline execution, improving system performance and reliability by 25%.

TECHNICAL SKILLS

Programming: Python, R, C++, SQL. **Frameworks:** PyTorch, TensorFlow, Hugging Face Transformers, FastAPI.
Machine Learning: Optimization modeling, relevance/ranking features, embeddings, model evaluation, deployment.
GenAI & LLMs: Transformer-based models, LLM fine-tuning, retrieval-augmented generation, multimodal modeling.
Experimentation: Golden datasets, A/B testing, lift analysis, telemetry, KPI definition, offline/online evaluation.
Cloud & MLOps: PostgreSQL, Data Warehousing, AWS (Bedrock, Athena, Redshift), GCP, Azure, Pinecone, Chroma.
Deep Learning: Neural Networks, CNNs, LSTMs, YOLO, Faster R-CNN, GANs, Time-Series Forecasting.

EDUCATION

Rutgers University, Master of Science in Data Science Sept 2022 - May 2024 | New Brunswick, NJ
Personal Website: <https://jayanthdasamantharao.github.io/>

Andhra University B.Tech in Electrical and Electronics Engineering 2017 - 2021 | Visakhapatnam, India

RESEARCH PROJECTS

Self-Driving Cars (Computer Vision): Built autonomous driving cars with vision pipeline using CNNs, YOLO, Faster R-CNN, achieving 95%+ object detection accuracy.

LLM-Powered Retrieval & Question Answering System (Production): Built and shipped LLM-powered retrieval and question answering services using embeddings, RAG, and evaluation-driven iteration and offline/online metric alignment, improving response quality, factual grounding, and production reliability.

Bayesian Classification: Implemented Bayesian logistic regression in PyMC3 to predict diabetes status, incorporating informative priors, achieving 0.85 F1-score.

For additional projects and code samples: <https://github.com/JayanthDasamantharao>